

HW2.

P1. Let P be an even integer, Since $W_t \sim N(0, t)$,

$$\begin{aligned}
 E[W_t^P] &= \int_{-\infty}^{+\infty} x^P \cdot \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t} dx \\
 &= \left[x^{P-1} \left(-\frac{t}{\sqrt{2\pi t}}\right) e^{-x^2/2t} \right]_{x=-\infty}^{x=+\infty} \\
 &\quad + (P-1) \int_{-\infty}^{+\infty} x^{P-2} \cdot \frac{t}{\sqrt{2\pi t}} e^{-x^2/2t} dx \rightarrow P-1 > 0 \\
 &= t(P-1) E[W_t^{P-2}] \rightarrow \text{as long as } P-3 > 0 \\
 &= t^2(P-1)(P-3) E[W_t^{P-4}] \rightarrow P-3 > 0 \\
 &= \dots \\
 &= t^{P/2} (P-1)(P-3) \dots 1 \cdot E[W_t^0] \\
 &= t^{P/2} (P-1)(P-3) \dots 1.
 \end{aligned}$$

P2. Yes. We verify the properties.

1. $X_0 = 0$

$$\begin{aligned}
 2. \lim_{t \rightarrow 0^+} X_t &= \lim_{t \rightarrow 0^+} t W_{1/t} \\
 &\stackrel{a.s.}{=} \lim_{t \rightarrow 0^+} t \cdot \sqrt{\frac{1}{t}} \\
 &= \lim_{t \rightarrow 0^+} \sqrt{t} \\
 &= 0 \Rightarrow \text{Continuous}
 \end{aligned}$$

$$\begin{aligned}
 3. X_t - X_s &= t W_{1/t} - s W_{1/s} \\
 &\quad \downarrow N(0, 1/t) \quad \downarrow N(0, 1/s) \\
 &\quad \underbrace{N(0, t) \quad N(0, s)}_{\downarrow} \\
 &\quad N(0, t-s)
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(X_t - X_s, X_s) &= \text{Cov}(X_t, X_s) - \text{Cov}(X_s, X_s) \\
 &= \text{Cov}(t W_{1/t}, s W_{1/s}) - \text{Var}(s W_{1/s}) \\
 &= ts \min\left(\frac{1}{t}, \frac{1}{s}\right) - s^2 \cdot \frac{1}{s} \\
 &= ts \cdot \frac{1}{s} - s = s - s = 0
 \end{aligned}$$

 $\Rightarrow$ stationary and independent increments

4. $X_t = X_t - X_0 \stackrel{d}{=} N(0, t)$

Thus X_s is a Brownian motion.

P3.

$$\begin{aligned}
 C_X(t, s) &= \text{Cov}(X_t, X_s) \\
 &= \text{Cov}(e^{-t} W_{e^{2t}}, e^{-s} W_{e^{2s}}) \\
 &= e^{-(t+s)} \text{Cov}(W_{e^{2t}}, W_{e^{2s}}) \\
 &= e^{-(t+s)} \min(e^{2t}, e^{2s}) \\
 &= e^{-(t+s)} e^{2 \min(s, t)} \\
 &= e^{-|t-s|}
 \end{aligned}$$

P4.

$$\begin{aligned}
 M_X(t) &= E[X_t] = E\left[\frac{W_{t+h} - W_t}{h}\right] \\
 &= \frac{1}{h} (E[W_{t+h}] - E[W_t]) = 0
 \end{aligned}$$

$$\begin{aligned}
 C_X(t, s) &= E[X_t X_s] - E[X_t] E[X_s]. \text{ let } s \leq t. \\
 &= E[X_t X_s] \\
 &= E\left[\frac{W_{t+h} - W_t}{h} \cdot \frac{W_{s+h} - W_s}{h}\right] \\
 &= \frac{1}{h^2} (E[W_{t+h} W_{s+h}] - E[W_{t+h} W_s] \\
 &\quad - E[W_t W_{s+h}] + E[W_t W_s]) \\
 &= \frac{1}{h^2} ((s+h) - s - \min(s+h, t) + s) \\
 &= \frac{1}{h^2} (s+h - \min(s+h, t))
 \end{aligned}$$

P5. Year Close S_j ΔS_j $\frac{\Delta S_j}{S_j}$

2000	\$4.595	0.055	0.0120
2001	\$4.650	0.140	0.0301
2002	\$4.790	1.170	0.2443
2003	\$5.960	0.885	0.1485
2004	\$6.845	2.065	0.3017
2005	\$8.910		

$$\begin{aligned}
 \alpha &= \frac{1}{5} \sum_{j=1}^5 \frac{\Delta S_j}{S_j} \approx 0.1473 = 14.7\% \text{ per year} \\
 \beta &= \sqrt{\text{Var}\left(\frac{\Delta S_j}{S_j}\right)} = \left(\frac{1}{4} \sum_{j=1}^5 \left(\frac{\Delta S_j}{S_j} - \alpha\right)^2\right)^{\frac{1}{2}} \\
 &\approx 0.1277 = 12.8\% \text{ per } \sqrt{\text{year}}
 \end{aligned}$$